



Figure 1: Software development and the cloud IDE

based collaborative software development. Let's also analyse a cloud based development facility that can be accessible from anywhere, and be used without much interfacing in the local developer system such as software installation, network boundary, etc. For example, let's only use a Web browser to access the cloud IDE for software development.

A cloud based IDE is a 'next gen' solution and still in the incubation stage, because the cloud based private workspace is yet to evolve fully. As of now, there still isn't any good Java-supporting cloud IDE available. For example, eXo provides facilities, but it lags in speed to some extent and doesn't support the IE browser.

We have reviewed various cloud IDEs, both commercial and open source, and have shortlisted cloud IDE tools for review, for the benefit of open source users. In this regard, we have considered the following standard expectations from a cloud IDE:

1. It should be lightweight and flexible enough for novices.
2. It should be able to highlight syntax.
3. It should have the auto completion facility.
4. It should have an internal compilation facility.
5. It should provide minimum debugging facilities.
6. It should support remote repository management for source code.
7. It should support cloud deployment.

In addition to this, proficient developers who have worked on various IDEs like Eclipse, JDeveloper, IntelliJ, Visual Studio, etc, would look for some add-on features like:

1. Support for screen design with a visual designer tool.
2. API or language help; for example, tooltip for a Java API.
3. Static code analysis to follow best practices in coding.
4. Coding convention and code formatting or styling; for example, the number of characters in a line, tab indentation, etc.

So let us review some of the popular open source cloud IDEs including Cloud9, codeanywhere, Eclipse Orion, Coderun Studio, Codebox, Codeiad, IDE One and Codenvy–Eclipse Che.

Please note that the tools discussed below are not ranked based on any parameters. The intention of this

article is to discuss various features, as well as the pros and cons of these tools based on our experience. The idea is not to influence decisions, since we are well aware that the development requirements decide what kind of cloud IDE or tools are chosen; so a general recommendation is not advisable.

Cloud9 - <https://c9.io/>

Cloud9 IDE is capable of handling projects from JS, HTML, PHP and Ruby. It supports development in JavaScript, GIT, Selenium, HTML/CSS, XML, XQuery, PHP, Ruby and Node.js. It can be used for Web application development. Cloud9 supports repositories like GIT, the distributed revision control tool Mercurial, and Apache Subversion. It also facilitates deployment, which can be done directly from Cloud9 to Joyent and Heroku. Its limitation is that it is not useful for Java application development or J2EE application development/handling.

Pros: Supports a variety of languages and various repository tools.

Cons: Can be used for simple application development but not for large or complex applications.

Codeanywhere - <https://codeanywhere.net>

This is a lightweight browser based IDE tool, which supports development of Web applications using HTML, Cascading Style Sheets (CSS), JavaScript, PHP, MySQL, and more. We can work with this tool to develop or test code from anywhere—even from mobile devices—since it is also available as a mobile application. Codeanywhere is supported in iOS, Android and Blackberry based mobile devices and tablets.

Pros: Lightweight and easy to use. Support for mobile devices.

Cons: More useful for Web page development and doesn't support a variety of programming languages.

Eclipse Orion - <https://wiki.eclipse.org/Orion>

This is the Eclipse version of Orion, which can be used as a local standalone (installable) IDE or an online IDE through a browser, but it only supports Chrome, FF, Safari and IE10. This is more for client side scripting like HTML, JS and CSS. Java syntax highlighting is supported during static code review, and doesn't integrate with the Java build and development environment. It has shell features with a very limited command facility. Eclipse Orion doesn't support the private cloud and hence is not suitable for secured application development. This is one of the lightweight cloud IDEs for software development in HTML/CSS and JavaScript.

Pros: Can be used as a standalone local installation tool or as a cloud IDE. It offers static code analysis. Shell feature provides facility for UNIX script development, which is a unique feature many cloud IDEs do not offer.

Cons: Doesn't support the private cloud/secure development, and doesn't support build/compilation of code or online testing.

Coderun Studio - www.coderun.com/studio/

This is another very simple but efficient cloud IDE, which